

AWS Account Setup & IAM Configuration Report

Objective

The objective of this week's activities was to establish a secure AWS environment for the project team by creating and configuring an AWS account, implementing identity and access management (IAM) best practices, enabling multi-factor authentication (MFA), monitoring account costs, and securely provisioning access for all team members. These activities provide a strong security foundation while ensuring effective collaboration throughout the internship.

Activities Performed

1. Created AWS Free Tier Account

- Successfully created an AWS Free Tier account for the internship environment.
- Verified the account setup and ensured access to the AWS Management Console.
- Confirmed that the account was eligible for AWS Free Tier services.

2. Enabled Multi-Factor Authentication (MFA) on the Root Account

- Navigated to the AWS IAM console and accessed the Security Credentials section.
- Configured Multi-Factor Authentication (MFA) for the root user using a supported authenticator application.
- Verified successful MFA registration by completing the authentication process.
- Confirmed that the root account is now protected with an additional layer of security.

3. Created IAM Users for Six Team Members

- Accessed the AWS Identity and Access Management (IAM) service.
- Created six individual IAM user accounts, one for each team member.
- Assigned unique usernames to facilitate user identification and accountability.
- Enabled AWS Management Console access for users requiring console login.

4. Applied Least-Privilege IAM Permissions

- Created and assigned IAM policies based on the principle of least privilege.
- Granted each team member only the permissions necessary to perform assigned tasks.
- Avoid the use of overly permissive administrative privileges unless specifically required.
- Verified that each user's permissions aligned with their intended responsibilities.

5. Configured AWS Budget and Billing Alarm

- Open the AWS Billing and Cost Management console.
- Created an AWS Budget with a monthly spending limit of **USD 10**.
- Configured budget notifications to alert the account owner before the spending limit is exceeded.
- Verified that budget monitoring was successfully enabled to support responsible cloud resource usage.

6. Shared IAM Credentials Securely

- Generated temporary login credentials for each IAM user.
- Shared credentials securely with each team member through approved communication channels.
- Instructed users to change their temporary passwords upon first login.
- Emphasized AWS security best practices, including safeguarding credentials and enabling MFA where applicable.

Outcome

The AWS environment was successfully configured with secure identity and access management practices. All team members received individual IAM accounts with appropriate permissions; the root account was protected using Multi-Factor Authentication, and a billing budget was established to monitor cloud expenditure. These configurations provide a secure and well-managed foundation for subsequent AWS Cloud Computing and AI practical activities.

Conclusion

During the first week of the AWS Cloud Computing and AI Training with Azubi Africa, the foundational AWS account and security configurations were successfully completed. Implementing IAM best practices, enabling MFA, configuring cost monitoring, and securely provisioning user access established in a secure, organized, and cost-conscious cloud environment. These activities ensure that future cloud deployments and practical exercises can be carried out securely while adhering to AWS security and governance best practices.